

Recap

De Morgan's Laws

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

$$\neg(\forall x P(x)) \equiv \exists x \neg P(x)$$

$$\neg(\exists x P(x)) \equiv \forall x \neg P(x)$$

CD-Equivalence

$$p \rightarrow q \equiv \neg p \vee q$$

1.5 Nested Quantifiers

If a predicate involves more than one variable we can quantify using multiple quantifiers. (Read left to right)

Note

Two copies of the same type of quantifier commute (commutativity)

$$\forall x \forall y P(x, y)$$

Is the same as:

$$\forall y \forall x P(x, y)$$

Any counterexample to one would be a counterexample to the other.

Similarly,

$$\exists x \exists y P(x, y) \equiv \exists y \exists x P(x, y)$$

This does not work for a mix of quantifiers

Example

On the domain of integers:

Let $Q(x)$ be $2x + y = 3$

We have given an already picked x , we can let $y = 3 - 2x$ so $2x + y = 2x + (3 - 2x) = 3$ and thus $\exists y P(x, y)$

Since we didn't use any properties of x

So:

$$\forall x \exists y P(x, y)$$

End example

On the other hand, no matter which y is picked there is some x so that $2x + y \neq 3$

This means $\forall x P(x, y)$ is **False** (no matter the y)

So:

$$\neg \exists y \forall x P(x, y)$$

In general as long as the domain is non-empty:

$$\exists x \forall y P(x, y) \rightarrow \forall y \exists x P(x, y)$$

Example 2

Let $R(x, y)$ be $x + y = x$

In this case $y = 0$ gives the example that

$$\exists y \forall x R(x, y)$$

But also showing:

$$\forall x \exists y R(x, y)$$

There is not generally a relation between

$$\forall x \exists y P(x, y)$$

And:

$$\forall y \exists x P(x, y)$$

Calculus Example

$$f(x)R \rightarrow R$$

is continuous at x_0 iff

$$\forall \varepsilon > 0, \exists \delta > 0 (|x - x_0| < \delta \rightarrow |f(x) - f(x_0)| < \varepsilon)$$

We have the tools to say what it takes to not be continuous at x_0

$$\neg (\forall \varepsilon > 0, \exists \delta > 0, \forall x (|x - x_0| < \delta \rightarrow |f(x) - f(x_0)| < \varepsilon))$$

$$\equiv \exists \varepsilon > 0, \forall \delta > 0, \exists x (|x - x_0| < \delta \wedge |f(x) - f(x_0)| > \varepsilon)$$

Aside

$$\neg(p \rightarrow q) \equiv \neg p \vee q \text{ conditional-disjunction equivalence}$$

$$\equiv \neg \neg p \wedge \neg q \text{ De Morgans}$$

$$\equiv p \wedge \neg q \text{ Double Negation}$$

1.6 Rules of Inference

Framework:

If the hypothesis $P_1, P_2, \dots, P_n$ give enough information to conclude q , we write:

$$\frac{p_1 \quad p_2 \quad \dots \quad p_n}{\therefore q}$$

This is the case if:

$$(p_1 \wedge p_2 \dots \wedge p_n) \rightarrow q$$

is a tautology.

For this to not be a tautology, there must be an assignment of variables so that $p_1 \wedge p_2 \dots \wedge p_n$ is **true** but q is **false**

Example

Can we conclude from $p \wedge q$ that $p \vee q$?

This asks is

$$p \wedge q$$

$$\therefore p \vee q \text{ i.e. } p \wedge q \rightarrow p \vee q$$

is a tautology

p	q	$p \wedge q$	$p \vee q$	$(p \wedge q) \rightarrow (p \vee q)$
T	T	T	T	T
T	F	F	T	T
F	T	F	T	T
F	F	F	F	T

We check if only Ts are left in the conclusion column

Alternatively

1. $p \wedge q$ hypothesis

2. $\therefore p$ simplification applied to 1.

3. $\therefore p \vee q$ addition on 2.

Simplification: we can ignore extraneous facts

Addition: we can disjoin anything to something already true

If the conclusion is not justified, the argument is a fallacy.

Example

“One of my students was a brilliant math guy or an NFL player”

He was an NFL player

$\therefore$ He was not a brilliant math guy

with

p “He was brilliant”

q “he was in NFL?”

Our argument is :

$p \vee q$

q

$\therefore \neg p$

But all hypotheses are true if p is T and q is T but then $\neg p$ is false